Q1 - 24 June - Shift 1

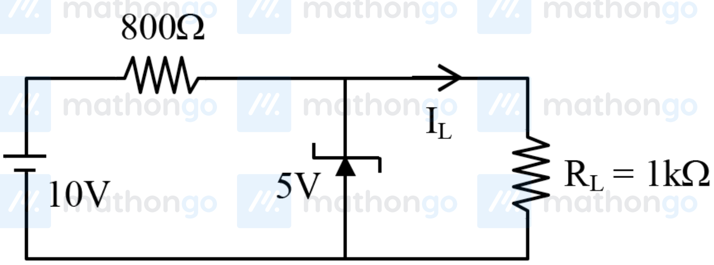
A transistor is used in common-emitter mode in an amplifier circuit. When a signal of 10 mV is added to the base-emitter voltage, the base current changes by 10 μA and the collector current changes by 1.5 mA. The load resistance is 5 kΩ. The voltage gain of the transistor will be _____ .

Space for your notes:

Q2 - 24 June - Shift 2

In the given circuit- the value of current I_L will be ______ mA.

(When $R_L = 1k\Omega$)



Space for your notes:

Q3 - 25 June - Shift 1

The photodiode is used to detect the optical signals. These diodes are preferably operated in reverse biased mode because.

(A) fractional change in majority carriers produce higher forward bias current

(B) fractional change in majority carriers produce higher reverse bias current

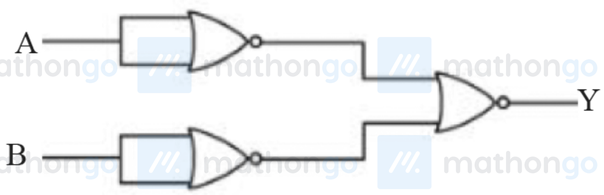
(C) fractional change in minority carriers produce higher forward bias current

(D) fractional change in minority carriers produce higher reverse bias current

Space for your notes:

Q4 - 25 June - Shift 2

Identify the logic operation performed by the given circuit :



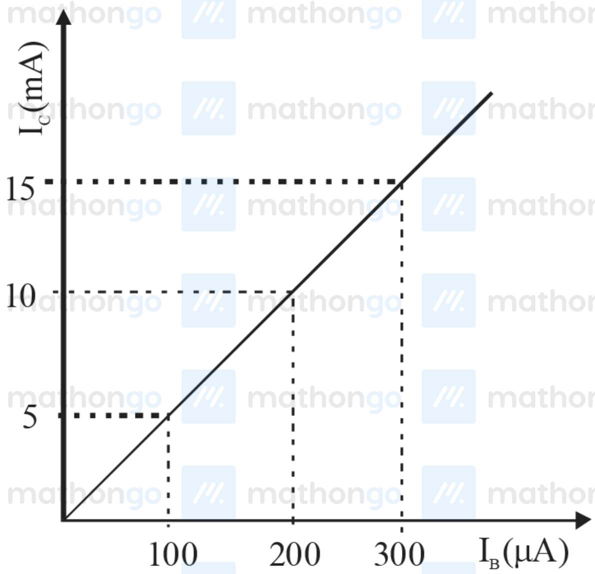
(A) AND gate (B) OR gate

(C) NOR gate (D) NAND gate

Space for your notes:

Q5 - 25 June - Shift 2

In an experiment of CE configuration of n-p-n transistor, the transfer characteristics are observed as given in figure.

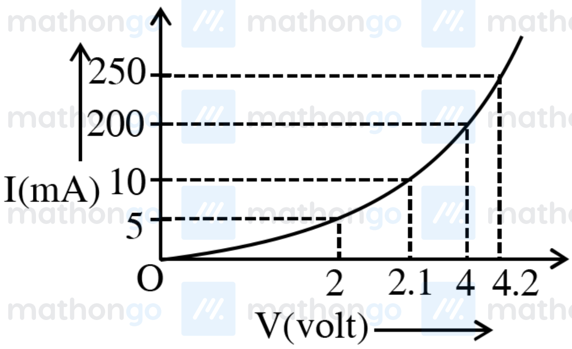


If the input resistance is 200Ω and output resistance is 60 Ω the voltage gain in this experiment will be ______

Space for your notes:

Q6 - 26 June - Shift 1

The I-V characteristics of a p-n junction diode in forward bias is shown in the figure. The ratio of dynamic resistance, corresponding to forward bias voltages of 2V and 4V respectively, is :

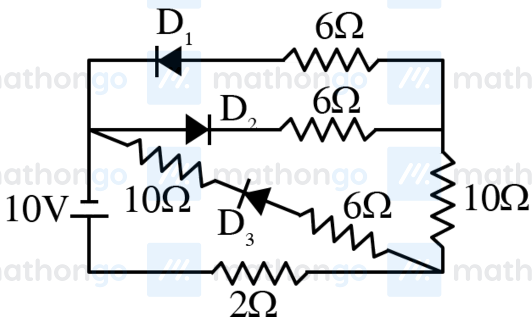


(A) 1 : 2
(B) 5 : 1
(C) 1 : 40
(D) 20 : 1

Space for your notes:

Q7 - 26 June - Shift 1

As per the given circuit, the value of current through the battery will be .... A.



Space for your notes:

Q8 - 26 June - Shift 2

The positive feedback is required by an amplifier to act an oscillator. The feedback here means :

(A) External input is necessary to sustain ac signal in output.

(B) A portion of the output power is returned back to the input.

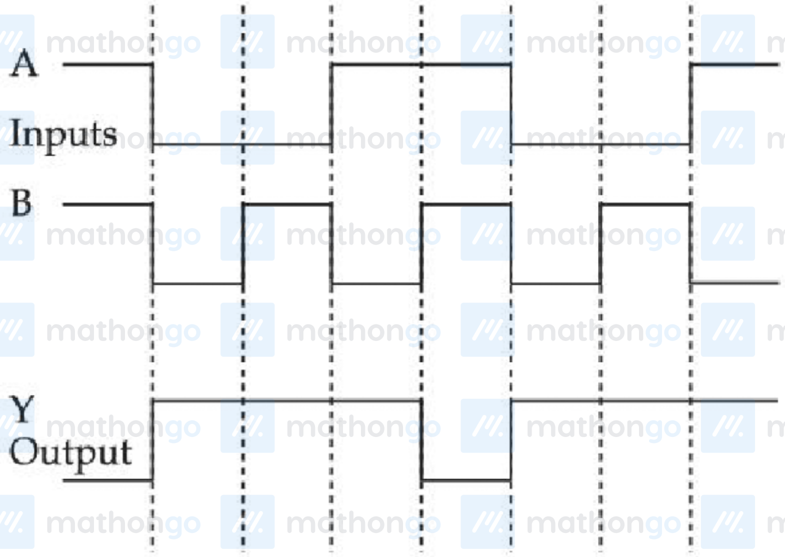
(C) Feedback can be achieved by LR network.

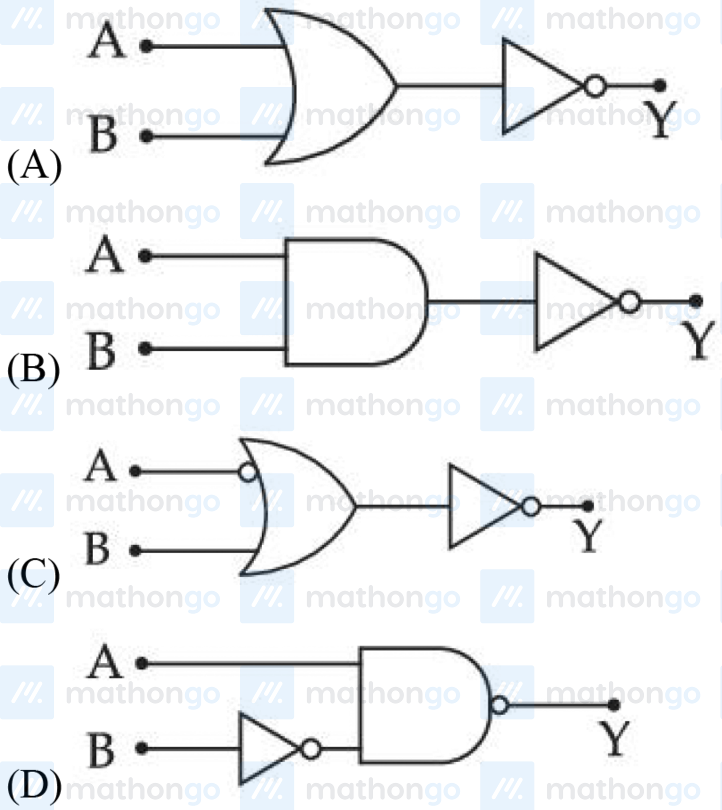
(D) The base-collector junction must be forward biased.

Space for your notes:

Q9 - 27 June - Shift 1

Identify the correct Logic Gate for the following output (Y) of two inputs A and B.





Space for your notes:

Q10 - 27 June - Shift 2

For a transistor to act as a switch, it must be operated in

(A) Active region

(B) Saturation state only

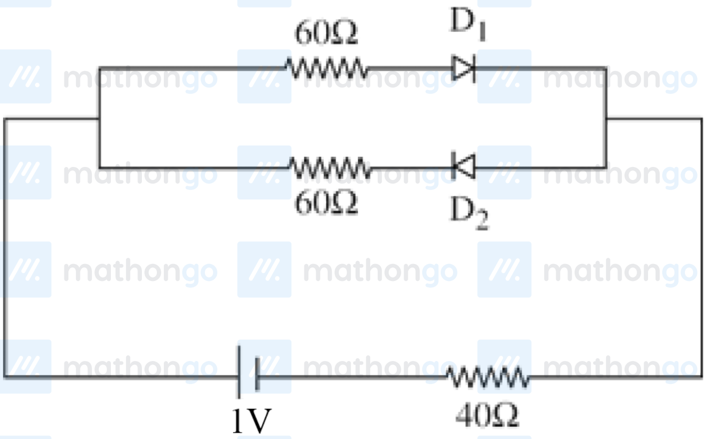
(C) Cut-off state only

(D) Saturation and cut-off state

Space for your notes:

Q11 - 27 June - Shift 2

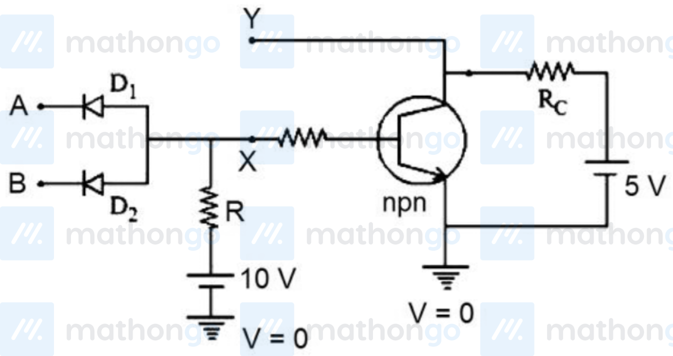
The cut-off voltage of the diodes (shown in figure) in forward bias is 0.6 V. The current through the resister of 40 Ω is _______ mA.



Space for your notes:

Q12 - 28 June - Shift 1

In the following circuit, the correct relation between output (Y) and inputs A and B will be :



(A) $Y = AB$ (B) $Y = A + B$

(C) $Y = \overline{AB}$ (D) $Y = \overline{A + B}$

Space for your notes:

Q13 - 28 June - Shift 1

For using a multimeter to identify diode from electrical components. choose the correct statement out of the following about the diode :

(A) It is two terminal device which conducts current in both directions.

(B) It is two terminal device which conducts current in one direction only

(C) It does not conduct current gives an initial deflection which decays to zero.

(D) It is three terminal device which conducts current in ne direction only between central terminal and either of the remaining two terminals

Space for your notes:

Q14 - 28 June - Shift 1

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : n-p-n transistor permits more current than a p-n-p transistor.

Reason R : Electrons have greater mobility as a charge carrier.

Choose the correct answer from the options given below :

(A) Both A and R true. and R is correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

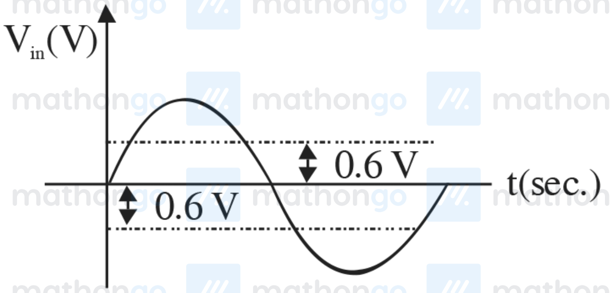
(C) A is true but R is false.

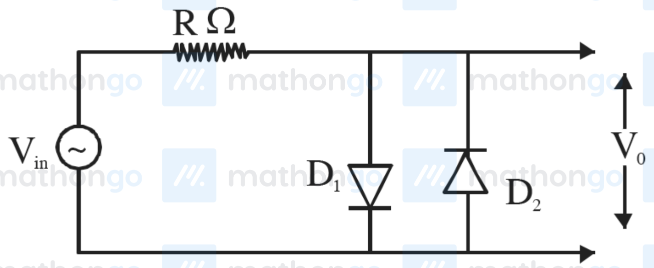
(D) A is false but R is true.

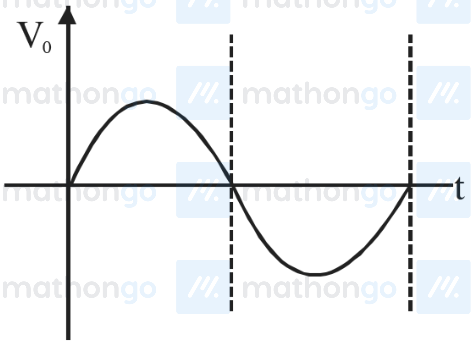
Space for your notes:

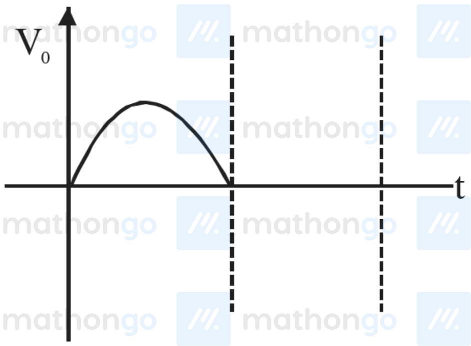
Q15 - 28 June - Shift 2

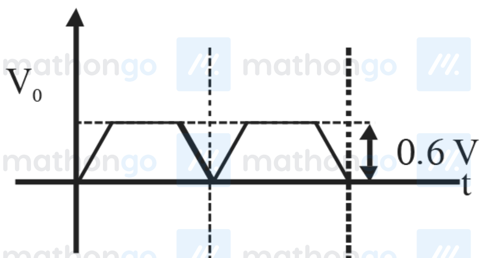
In the given circuit the input voltage V_{in} is shown in figure. The cut–in voltage of p–n junction diode (D_1 or D_2) is 0.6 V. Which of the following output voltage (V_0) waveform across the diode is correct?

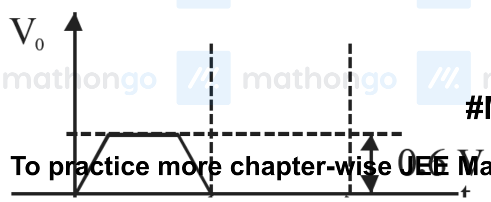




(A)


(B)


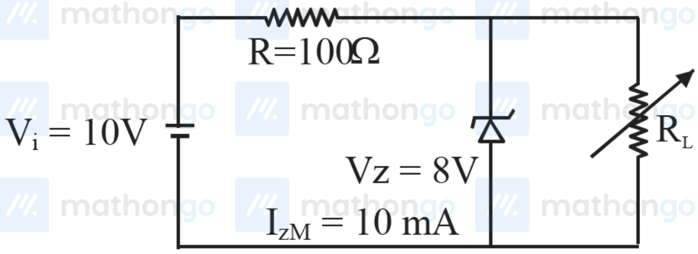
(C)


(D)


Space for your notes:

Q16 - 28 June - Shift 2

A Zener of breakdown voltage $V_Z = 8V$ and maximum zener current, $I_{ZM} = 10$ mA is subjected to an input voltage $V_i = 10V$ with series resistance $R = 100\Omega$. In the given circuit R_L represents the variable load resistance. The ratio of maximum and minimum value of R_L is ______



Space for your notes:

Q17 - 29 June - Shift 1

A transistor is used in an amplifier circuit in common emitter mode. If the base current changes by 100 μA, it brings a change of 10 mA in collector current. If the load resistance is 2 kΩ and input resistance is 1 kΩ, the value of power gain is $x \times 10^4$. The value of x is ______.

Space for your notes:

Q18 - 29 June - Shift 2

A potential barrier of 0.4 V exists across a p-n junction. An electron enters the junction from the n-side with a speed of $6.0 \times 10^5\ ms^{-1}$. The speed with which electron enters the p side will be $\frac{x}{3} \times 10^5\ ms^{-1}$ the value of x is ______.

(Given mass of electron = 9×10^{-31} kg, charge on electron = 1.6×10^{-19}C.)

Space for your notes:

Answer Key

Q1 (750)	**Q2** (5)	**Q3** (D)	**Q4** (A)
Q5 (15)	**Q6** (B)	**Q7** (1)	**Q8** (B)
Q9 (B)	**Q10** (D)	**Q11** (4)	**Q12** (C)
Q13 (B)	**Q14** (A)	**Q15** (D)	**Q16** (2)
Q17 (2)	**Q18** (14)		

Q1 (750)

$$r_i = \frac{10\text{mV}}{10\mu\text{A}} = 10^3\,\Omega$$

$$\beta = \frac{1.5\text{mA}}{10\mu\text{A}} = 150$$

$$A_V = \left(\frac{R_0}{r_i}\right)\beta = \left(\frac{5000}{1000}\right)\times 150 = 750$$

Q2 (5)

$$I_L = \frac{5}{1000} = 5\text{mA}$$

Q3 (D)

$$v \propto \frac{Z}{n} \propto Z \quad (n = \text{constant})$$

$$\Rightarrow \frac{v_{He^+}}{v_H} = \frac{Z_{He^+}}{Z_H} = \frac{2}{1}$$

Q4 (A)

$$= \left[\overline{\left[\overline{A+A}\right] + \left[\overline{B+B}\right]}\right]$$

$$Y = \overline{\overline{A} + \overline{B}}\ (\text{D' MORGAN LAW})$$

$$Y = AB$$

Q5 (15)

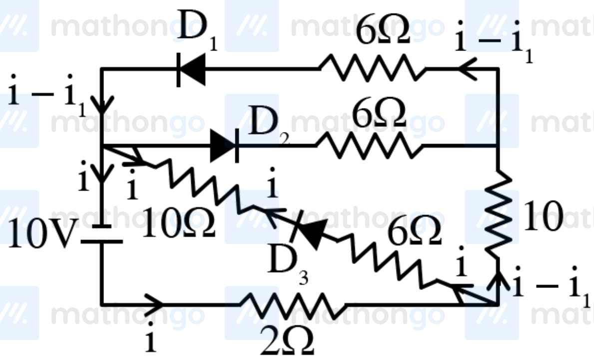
$$\text{Voltage Gain} = \frac{I_C}{I_B} \times \frac{R_0}{R_I} = \frac{10 \times 10^{-3}}{200 \times 10^{-6}} \times \frac{60}{200} = 15$$

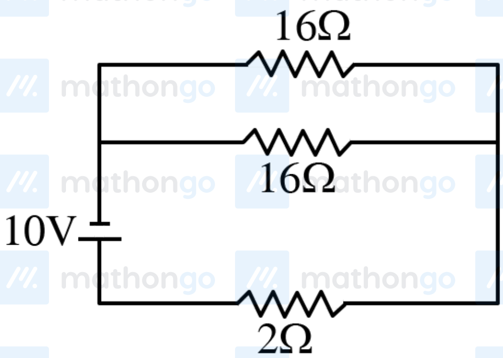
Q6 (B)

$$R = \frac{\Delta V}{\Delta i}$$

$$\frac{R_1}{R_2} = \frac{\Delta v_1}{\Delta v_2} \frac{\Delta i_2}{\Delta i_1} = \frac{0.1}{0.2} \times \frac{50}{5} = 5$$

Q7 (1)





$V = IR_{net}$

$10 = I \times 10$

$I = 1A$

Ans. 1

Q8 (B)

When the amplifier connects with positive feedback, it acts as the oscillator the feedback here is positive feedback which means some amount of voltage is given to the input.

Q9 (B)

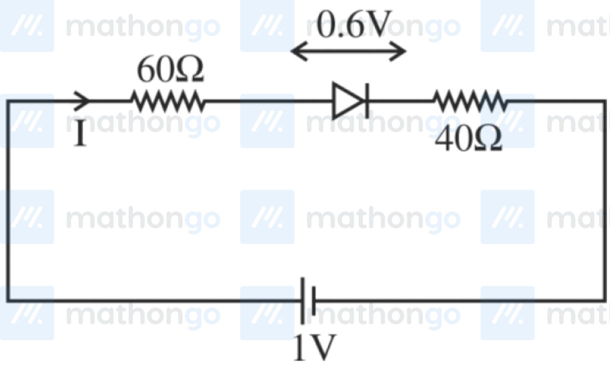
A	B	Y
1	1	0
0	0	1
0	1	1
1	0	1
1	1	0
0	0	1
0	1	1
1	0	1
NAND Gate		

$$Y = \overline{A.B}$$

Q10 (D)

Transistor act as a switch in saturation and cut of region.

Q11 (4)



$$1 - I(60) - 0.6 - I(40) = 0$$

$$\frac{0.4}{100} = I$$

$$I = 4\text{ mA}$$

Q12 (C)

This is NAND gate

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	0

Q13 (B)

In forward bias diode conducts

In revers bias it does not conducts.

Q14 (A)

Theory

Q15 (D)

In +ve half cycle

$D_1 \rightarrow$ F.B.; $\quad D_2 \rightarrow$ R.B.

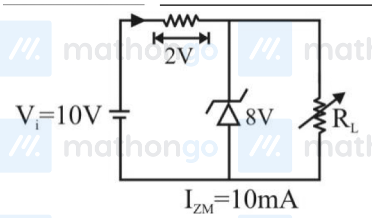
$0 - 0.6$ V

V_{out} same as V_{in}

In –ve half cycle

$D_2 \rightarrow$ F.B.; $\quad D_1 \rightarrow$ R.B.

Q16 (2)



$I = \frac{2}{100} = 20\text{mA}$ $V_L = I_L R_L$ $8 = 10 \times 10^{-3} \times R_{L_{max}}$ $\frac{4}{5} \times 10^3 = R_{L_{max}}$ $\boxed{800 = R_{L_{max}}}$	$I = I_Z + I_L$ $I_L = 10$ mA If $\boxed{I_Z = 0}$ $I_{L_{max}} = 20\text{mA}$ $V_L = I_{L_{max}} \times R_{L_{min}}$ $\frac{8}{20} \times 10^3 = R_{L_{min}}$ $\boxed{400 = R_{L_{min}}}$

$$\frac{R_{L_{max}}}{R_{L_{min}}} = \frac{800}{400} = 2$$

Q17 (2)

$\Delta i_B = 100\ \mu A$ $\beta = \frac{\Delta i_C}{\Delta i_B}$

$\Delta i_C = 10\ mA$

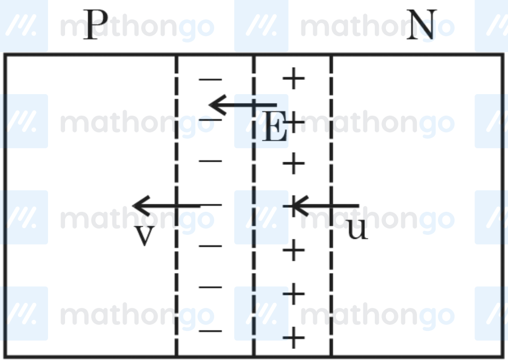
$\text{power} = \beta^2 \times \frac{R_0}{R_{in}}$

$\text{Power} = \left(\frac{10}{0.1}\right)^2 \times \frac{2}{1}$

$\text{Power} = 100 \times 100 \times 2$

$\text{Gain} = 2 \times 10^4$

Q18 (14)



Work done by Electric field = $K_f - K_i$

$$\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = -1.6 - 10^{-19} \times 0.4$$

$$\frac{1}{2}9 \times 10^{-31}(v^2 - u^2) = -0.64 \times 10^{-19}$$

$$u^2 - v^2 = \frac{2 \times 0.64 \times 10^{12}}{9}$$

$$v^2 = \left(36 - \frac{128}{9}\right) \times 10^{10}$$

$$v = \frac{14}{3} \times 10^5 \, m/s$$

$$x = 14$$